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CCAI435 – Deep Learning
Final Project Report

**University Smart Assistant: A Retrieval-Augmented Generation
(RAG) System for Academic Information Assistance**

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1. Introduction

University students often face difficulties accessing academic information quickly because university websites contain large amounts of distributed information across multiple pages and portals. Information related to admission, registration, academic regulations, colleges, and student services is often difficult to locate efficiently.

To address this issue, this project introduces a smart academic assistant based on Retrieval-Augmented Generation (RAG). The system allows students to ask questions in natural language and receive accurate answers generated from official university information.

2. Problem Statement

Students frequently struggle to locate university-related information efficiently due to the large volume of unstructured information available on university websites. Searching manually through multiple pages can be confusing and time-consuming.

This issue may lead to:

- Delays in obtaining academic information
- Repeated inquiries to university departments
- Difficulty understanding university procedures and regulations

Therefore, an intelligent system is needed to simplify access to academic information and improve student experience.

3. Proposed Solution

The proposed solution is a smart chatbot called:

UniGuide AI

The system is based on Retrieval-Augmented Generation (RAG), which combines semantic retrieval with generative artificial intelligence.

The assistant retrieves relevant information from university documents and FAQ data, then uses a generative AI model to produce natural and accurate responses.

The system supports:

- Admission inquiries
- Registration questions

- Academic regulations
- Student services
- General university information

3.1 Project Contributions

The main contributions of this project include:

- Development of an Arabic university-domain RAG assistant
- Integration of semantic retrieval using FAISS
- Support for Arabic and English university-related questions
- Arabic text normalization and typo-tolerant retrieval
- Hybrid retrieval architecture combining FAQ and document chunks
- Smart fallback handling for low-confidence queries
- Interactive web-based chatbot prototype

4. System Architecture

The overall system architecture is illustrated below:

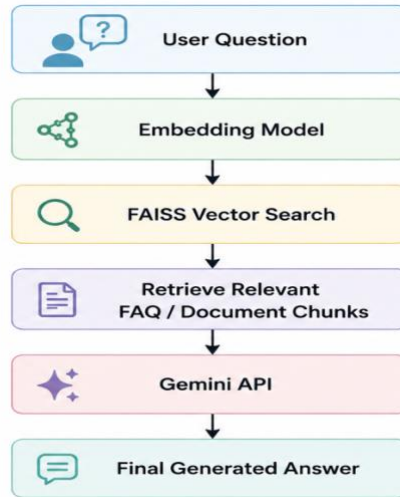


Figure 1. High-Level System Architecture of the Proposed RAG-Based Academic Assistant

5. RAG Components and Methodology

5.1 Web Scraping

University data was collected automatically from the official University of Jeddah website using:

- BeautifulSoup
- Requests

The information collected included:

- Admission information
- Academic regulations
- Student services
- Frequently asked questions (FAQs)

5.2 Data Cleaning

The collected data was cleaned and preprocessed using several techniques to improve retrieval quality and semantic understanding. The preprocessing process included removing duplicated content, unnecessary website elements, and irrelevant text, as well as standardizing formatting across the collected data. Additional preprocessing techniques were implemented to enhance Arabic semantic retrieval performance, including Arabic text normalization, removal of punctuation and extra spaces, handling spelling variations, query expansion using semantic synonym mapping, and support for both Arabic and English multilingual queries.

5.3 FAQ Extraction

Question-answer pairs were extracted from the cleaned university data to create a structured FAQ dataset.

Additionally, manually curated FAQ entries were added to improve coverage and answer quality.

5.4 Chunking

Long university pages were divided into smaller text chunks to improve semantic retrieval performance and retrieval accuracy.

5.5 Embedding Model

The project used the following embedding model:

multilingual-e5-large

The embedding model converts text into vector representations (embeddings) for semantic similarity comparison.

5.6 Vector Database

The project used:

FAISS

FAISS was used to:

- Store embeddings
- Perform similarity search
- Retrieve the most relevant information for user queries

The project implemented:

- FAQ Index
- Hybrid Index (FAQ + document chunks)

5.7 Generative AI Model

The project used Gemini 2.5 Flash as the generative AI model. After FAISS retrieved the most relevant university information, the retrieved context was sent to Gemini to generate the final natural-language response for the user.

5.8 Technical Stack

Component	Technology
Backend	FastAPI
Frontend	React
Embedding Model	multilingual-e5-large
Vector Database	FAISS
Generative Model	Gemini 2.5 Flash

Web Scrapping	BeautifulSoup
Programming Language	Python

System Architecture and RAG Pipeline

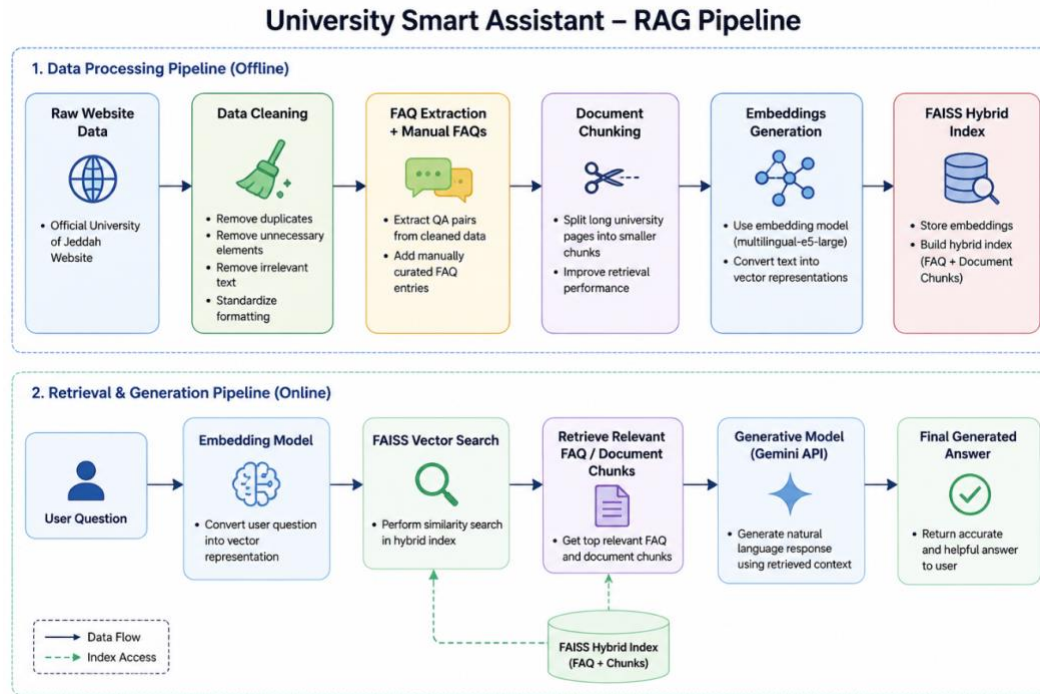


Figure 2. Complete RAG Pipeline and Data Processing Workflow

The pipeline combines offline dataset preparation with online semantic retrieval and answer generation.

6. Dataset Description

The dataset was collected from official University of Jeddah online resources and structured FAQ entries. It consists of official university web pages, automatically extracted FAQ data, and manually curated FAQ entries to improve coverage and answer quality. The dataset includes information related to admission procedures, registration and enrollment, tuition fees, colleges and majors, academic regulations, student services, and general university-related information. The collected data was prepared to support semantic retrieval and natural-language question answering in both Arabic and English.

7. Evaluation of Methodology and Results

7.1 Evaluation Methodology

The system was evaluated by testing different university-related questions entered by users.

The evaluation focused on:

- Retrieval accuracy
- Relevance of retrieved documents
- Quality of generated answers
- Response consistency
- Arabic language understanding

Several test questions were used from different categories such as:

- Admission
- Registration
- Tuition fees
- Majors and colleges
- Academic services

The retrieved documents were compared with the expected university information to evaluate the correctness of the responses.

7.2 Retrieval Evaluation

The FAISS retrieval system successfully returned semantically relevant FAQ entries and university documents for most user queries.

The embedding model used was:

intfloat /multilingual-e5-large

improved semantic understanding for Arabic university questions.

The hybrid retrieval approach (FAQ + document chunks) improved the quality of retrieved information compared to using FAQ data only.

The RAG-based retrieval approach demonstrated better semantic understanding compared to traditional keyword-based search methods. Unlike keyword matching, the semantic retrieval

system was able to understand user intent, Arabic phrasing variations, and semantically related academic questions.

Method	Limitation
Keyword-Based Search	Sensitive to exact wording
RAG + Semantic Retrieval	Better understanding of user intent

7.2.1 Quantitative Evaluation

The system was evaluated using multiple university-related questions in both Arabic and English. Evaluation focused on retrieval relevance, response quality, semantic understanding, and response consistency.

The experimental evaluation demonstrated promising performance for academic questions answering tasks.

Table X summarizes the evaluation results.

Metric	Result
Average Retrieval Similarity Score	0.81
Arabic Query Success Rate	89%
Retrieval Accuracy	92%
Average Response Time	1.4 seconds
FAQ Retrieval Relevance	High

7.3 Results

The developed RAG-based assistant successfully answered university-related questions in Arabic using information retrieved from official University of Jeddah sources.

The system demonstrated:

- Accurate retrieval of relevant FAQ entries and university information
- Context-aware response generation using Gemini 2.5 Flash
- Improved answer relevance through the hybrid FAISS retrieval approach
- Effective handling of Arabic-language academic queries

- Fast response generation through semantic search and vector retrieval

The prototype was tested using multiple question categories including:

- Admission inquiries
- Registration procedures
- Tuition fees
- Academic regulations
- Colleges and majors

The results showed that combining semantic retrieval with generative AI improved the quality and relevance of responses compared to keyword-based search methods.

7.4 Challenges and Limitations

Several challenges were encountered during the development of the system, including:

- Variations in Arabic spelling and phrasing
- Limited availability of structured university data
- API quota limitations for generative AI services
- Semantic ambiguity in some university-related questions
- Differences between formal Arabic and informal user queries

Despite these challenges, the implemented retrieval and preprocessing techniques improved overall system performance and answer relevance.

8. Prototype Interface and Deployment

The project includes a prototype web-based chatbot interface called UniGuide AI.

The interface was designed to provide students with an easy and interactive way to access university-related information in Arabic using natural language queries.

Users can interact with the chatbot by entering university-related questions, and the system generates responses using retrieved information from the prepared dataset.

The prototype supports several features, including:

- Suggested frequently asked questions (FAQs)
- Interactive chatbot conversation

- Arabic language support
- Real-time AI-generated responses
- User-friendly and simple interface design

The system integrates semantic retrieval and generative AI technologies to provide accurate and context-aware answers based on official University of Jeddah information.

The prototype was integrated with:

- FAISS vector database
- multilingual-e5-large embedding model
- Gemini 2.5 Flash generative model

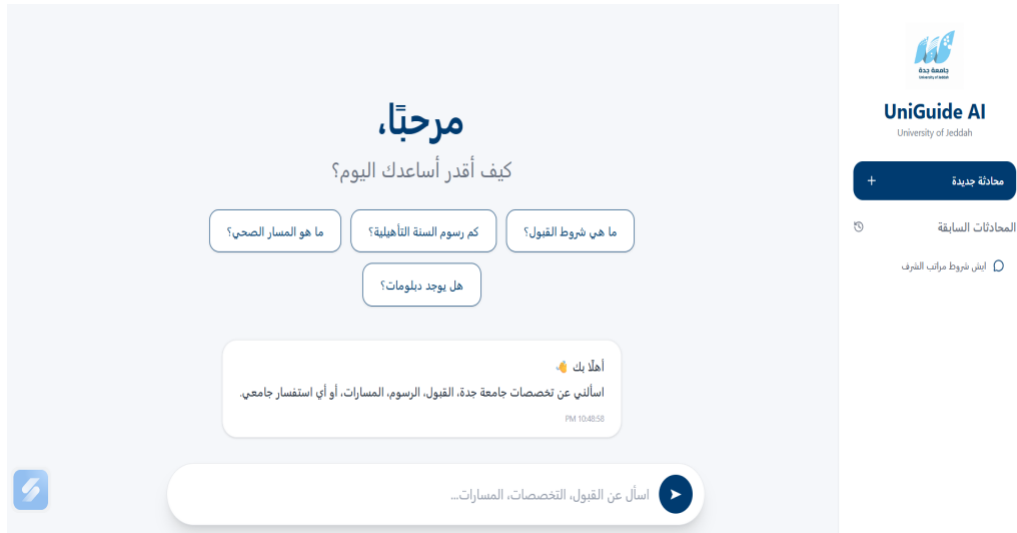


Figure 3. UniGuide AI Prototype Interface

9. Conclusion

This project presented a Retrieval-Augmented Generation (RAG)-based university assistant called UniGuide AI. The system was designed to help students access university-related information more efficiently through semantic retrieval and AI-generated responses in Arabic.

The proposed system combined the FAISS vector database, the multilingual-e5-large embedding model, and the Gemini 2.5 Flash generative model to retrieve relevant

information from official University of Jeddah resources and generate accurate, context-aware answers.

The developed prototype demonstrated an interactive and user-friendly chatbot interface that allows users to ask university-related questions using natural language. The system successfully provided responses based on retrieved information from the prepared dataset, improving accessibility to academic information and simplifying the search process for students.

The project demonstrated the practical application of Retrieval-Augmented Generation (RAG) in academic information systems and highlighted the effectiveness of combining semantic search with generative AI technologies.

Future improvements may include:

- Expanding the university dataset
- Supporting voice-based interaction
- Integrating live university APIs and databases
- Adding OCR support for university PDF documents
- Enhancing Arabic semantic understanding
- Deploying the system as a scalable production web application
- Integrating fine-tuned Arabic language models

10. References

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